1. Identification

Name of substance or mixture (trade name): Hydrous Ethanol EHC Fuel
Main recommended uses for the substance or mixture: Automotive fuel.
Internal code of identification of the substance or mixture: N/A

Production Units					
Company Name	Inpasa agroindustrial S/A	Inpasa agroindustrial S/A	Inpasa agroindustrial S/A	Inpasa agroindustrial S/A	Inpasa agroindustrial S/A
Address	ROD BR 163 KM 817 CEP 78.559-899	ROD BR 163 KM 603 CEP 78.450-000	ROD BR 163 KM 243 Zona Rural CEP 79.849-899	ROD BR 060 KM 417 Zona Rural CEP 79170-000	RODOVIA BR 230 KM 020 Zona Rural CEP 65800-000
Complement	Sinop – MT	Nova Mutum - MT	Dourados - MS	Sidrolândia - MS	Balsas - MA

Contact phone number: + 55 (66) 3531-5494 / 0800 800 9595

2. Identificação de perigos

Product Hazard Classification:

- Flammable liquids – Category 2
- Serious eye injuries/eye irritation – Category 2A
- Mutagenicidade em células germinativas – Categoria 1B
- Reproductive toxicity – Category 1A
- Specific end-organ toxicity – Single exposure – Category 3
- Specific Target Organ Toxicity – Repeated Exposure – Category 1 and 2
-

Classification system used: ABNT-NBR 14725-2:2009 standard – corrected version 2:2010.

Globally Harmonized System for the Classification and Labeling of Chemicals, UN.

Other hazards that do not result in a rating: Vapors can form explosive mixtures with air.

3. Composition and ingredient information

Product type: Substance.

Common chemical name or technical name: Ethanol.

Synonym: Ethyl alcohol; hydrated alcohol, AEHC.

Chemical nature: Hydrocarbons.

CAS Registration Number: 64-17-5 - min. 92.6 – 93.8% (w/w)

Impurities contributing to the hazard: Gasoline (CAS: 8006-61-9) - max. 30 mL/L (w/w)

4. First aid measures

Inhalation:	Remove the victim to a ventilated place and keep him at rest in a position that does not hinder breathing. If you feel unwell, contact a POISON CONTROL CENTER or a doctor. Take this MSDS.
Skin contact:	Wash exposed skin with enough water to remove the material. In case of skin irritation: Consult a doctor. Take this MSDS.
Contact with eyes:	Rinse thoroughly with water for several minutes. In the case of contact lens wear, remove them if it is easy. Continue rinsing. If eye irritation persists: consult a doctor. Take this MSDS.
Ingestion:	Do not induce vomiting. Never give something orally to an unconscious person. Rinse the victim's mouth with plenty of water. If you feel unwell, contact a POISON CONTROL CENTER or a doctor. Take this MSDS.

Most important symptoms and effects, acute or late:	It causes irritation to the skin with redness and dryness, and to the eyes with redness, pain and tearing. It can cause irritation of the airways with coughing, sneezing and shortness of breath. It can cause drowsiness, vertigo and headache. It can cause nausea and vomiting if ingested. It can cause damage to the central nervous system and liver through repeated and prolonged exposure. It can be fatal if aspirated if it penetrates the airways, resulting in chemical pneumonitis.
Notes to Doctor:	Avoid contact with the product when helping the victim. If necessary, symptomatic treatment should include, above all, supportive measures such as correction of hydroelectrolytic and metabolic disorders, as well as respiratory assistance. In case of contact with the skin, do not rub the affected area.

5. Firefighting measures

Means of extinguishing:	Suitable: Compatible with chemical powder, alcohol-resistant foam, carbon dioxide (CO_2) and water mist. Not recommended: Water jets. Water directly onto the burning liquid.
Hazards specific to the mixture or substance:	Highly flammable product. Containers can explode when heated. When heated, it can release toxic and irritating fumes. Risk of explosion indoors.
Fire fighting team protection measures:	Highly flammable product. Containers can explode when heated. When heated, it can release toxic and irritating fumes. Risk of explosion indoors.

6. Control measures for spillage or leakage

PERSONAL PRECAUTIONS

For non-emergency services personnel:	Isolate the leak from ignition sources. Prevent sparks or flames. Do not smoke. Evacuate the area within a radius of 300 meters. Do not touch damaged containers or spilled material without wearing proper clothing. Avoid inhalation, contact with eyes and skin. Wear personal protective equipment as described in section 8.
For emergency service personnel:	Use complete PPE, with side protection goggles, PVC protective gloves, safety shoes and waterproof protective clothing. In case of large leaks, where exposure is great, it is recommended to use a protective mask with an organic vapor filter.
Environmental precautions:	Prevent the spilled product from reaching waterways and sewer systems.
Methods and materials for containment and cleaning:	Use water mist or vapor suppressant foam to reduce the dispersion of vapors. Use natural or spill containment barriers. Collect the spilled product and place it in proper containers. Adsorb the remaining product with dry sand, soil, vermiculite, or any other inert material. Place the adsorbed material in appropriate containers and remove them to a safe place. For final disposal, proceed in accordance with Section 13 of this MSDS.
Diferenças na ação de grandes e pequenos vazamentos:	Differences in the action of large and small leaks:

7. Handling and storage

APPROPRIATE TECHNICAL MEASURES FOR HANDLING

Precautions for safe handling:	Handle in a ventilated area or with a general local ventilation/exhaust system. Avoid formation of vapors or mists. Avoid exposure to the product. Avoid contact with incompatible materials. Wear personal protective equipment as described in section 8.

Hygiene measures: Wash your hands and face thoroughly after handling and before eating, drinking, smoking, or going to the bathroom. Contaminated clothing should be changed and washed before reuse. Remove contaminated clothing and protective equipment before entering feeding areas.

Conditions for safe storage, including any incompatibilities

Fire and explosion prevention: Keep away from heat, sparks, open flames and hot surfaces. — Do not smoke. Keep container tightly closed. Ground the container and receiver during transfers. Use only non-sparking tools. Avoid the build-up of electrostatic charges. Use explosion-proof electrical, ventilation and lighting equipment.

Suitable conditions: Keep the product in a cool, dry and well-ventilated place, away from sources of heat and ignition. The storage location must have a containment basin to retain the product in case of a leak. Keep containers tightly closed and properly identified. The storage location must have an impermeable floor, free of combustible materials and a containment dam to retain in case of a leak. Keep away from incompatible materials. It is not necessary to add stabilizers and antioxidants to ensure the durability of the product.

Packaging materials: Similar to original packaging.

8. Exposure control and personal protection

Control parameters

Occupational exposure limits:

Component	TLV – TWA (ACGIH, 2012)	TLV – STEL (ACGIH, 2012)	LT (NR-15, 1978)
Ethanol	-	1000 ppm	780 ppm
Gasoline	300 ppm	500 ppm	-

Biological indicators:	Not established.
Engineering control measures:	Provide mechanical ventilation and a direct exhaust system to the outside environment. These measures help reduce exposure to the product. Keep atmospheric concentrations of the product's constituents below the indicated occupational exposure limits.

Personal protective measures

Eye protection:	Side protection glasses.
Skin and body protection:	For spill or leak control measures, use PVC protective gloves and appropriate protective clothing made of waterproof material. For handling, use PVC protective gloves, closed safety shoes and protective clothing against Flash Fire (FR).
Respiratory protection:	The use of a respirator with an organic vapor filter is recommended for average exposures above half the TLV-TWA. In cases where the exposure exceeds 3 times the TLV-TWA value, use a self-contained breathing apparatus (SCBA) with air supply, with a full facepiece, operated in positive pressure mode. Follow the guidelines of the Respiratory Prevention Program (PPR), 4th ed. São Paulo: Fundacentro, 2016.
Thermal hazards:	It does not present thermal hazards.

9. Physical and chemical properties***Aspecto:***

- **Physical state:** Liquid.
- **Form:** Clear.
- **Color:** Colorless.
- **Odor**: Characteristic.
- **Odor limit:** 180 ppm.
- **pH:** 6.0 - 8.0.
- **Melting point / freezing point:** -117ºC.
- **Initial boiling point:** 77°C.
- **Flash point:** 15ºC (closed cup).
- **Evaporation rate:** Not available.
- **Flammability (solid; gas):** Flammable product.

- **Lower flammability or explosive limit:** 3.3%.
- **Upper flammability or explosive limit:** 19 %.
- **Vapor pressure:** 5.8 kPa at 20ºC.
- **Vapor density:** 1.6 (air = 1).
- **Relative density:** 0.8 (water at 4°C = 1).
- **Solubility(ies):** Miscible in water, ethyl ether, acetone and chloroform. Soluble in benzene.
- **Partition coefficient n-octanol/water:** Log kow: -0.32.
- **Autoignition temperature:** 363º.
- **Decomposition temperature:** Not available.
- **Viscosity:** 1.20 cP at 20ºC.
- **Other information:** Not applicable.

10. Stability and Reactivity

Chemical stability and reactivity: Stable under normal handling and storage conditions. Does not undergo polymerization.

Possibility of hazardous reactions: Reacts violently with strong oxidants such as nitric acid, silver nitrate, mercuric nitrate or magnesium perchlorate with risk of fire and explosion.

Conditions to avoid: High temperatures. Sources of ignition and contact with incompatible materials.

Incompatible materials: Nitric acid, perchloric acid, permanganic acid, chromic anhydride, acetyl chloride, calcium hypochlorite, silver nitrate, mercuric nitrate, hydrogen peroxide, bromine pentafluoride, perchlorates, silver oxide, ammonia and oxidants in general.

Hazardous decomposition products: On combustion, releases toxic and irritating vapors.

11. Toxicological information

Acute toxicity: Product not classified as acutely toxic by oral route.

LD50 (oral, rats): 7060 mg/kg

Skin corrosion/irritation: Causes skin irritation with redness and dryness.

Serious eye damage/eye irritation: Causes eye irritation with redness, pain and tearing. Repeated eye contact may cause chronic conjunctivitis.

Respiratory or skin sensitization: The product is not expected to cause respiratory or skin sensitization.

Germ cell mutagenicity: May cause genetic defects if ingested.

Information regarding:

- **Ethanol:** Positive results for in vivo mutagenicity tests involving mammalian germ and somatic cells.

Carcinogenicity: The product is not expected to be carcinogenic.

Information regarding:

- **Ethanol:**

Not classified as carcinogenic to humans (IARC).

- **Gasoline:**

Possible carcinogen to humans (IARC – Group 2B).

Reproductive toxicity: May harm fertility or the fetus if ingested.

Information regarding:

Ethanol:

Can cause miscarriages, as well as birth defects and other developmental problems.

Specific target organ toxicity/single exposure: May cause central nervous system depression with dizziness, drowsiness, vertigo, headache, motor incoordination and loss of consciousness. May cause irritation of the respiratory and gastrointestinal tract with cough, sore throat, nausea, burning sensation, abdominal pain and diarrhea.

Specific target organ toxicity/repeated exposure: May cause central nervous system damage with tremors, irritability, loss of concentration and mental confusion through prolonged or repeated exposure. Cause liver damage with accumulation of fat in the liver and cirrhosis in case of chronic exposure by ingestion. May cause dry skin after repeated contact.

Aspiration hazard: The product is not expected to present an aspiration hazard.

Other information: Not available.

12. Ecological information

Environmental effects, behaviors and product impacts

Ecotoxicity: Not classified as hazardous to aquatic organisms.

CL50 (Salmo gairdnerii, 96h): 13000 mg/L.

Persistence and degradability: Rapid degradation and low persistence are expected.

Bioaccumulative potential: It has low bioaccumulative potential in aquatic organisms. BCF: 3. Log kow: -0.32.

Mobility in soil: High.

Other adverse effects: No other environmental effects are known for this product.

13. Considerations on final destination

Recommended methods for treatment and disposal applied to:

- **Product:** Must be disposed of as hazardous waste in accordance with local legislation. Treatment and disposal must be assessed specifically for each product. Federal, state and municipal legislation must be consulted, including: Law No. 12,305 of August 2, 2010 (National Solid Waste Policy).
- **Product remains:** Keep product remains in their original packaging and properly closed. Disposal must be carried out as established for the product.
- **Used packaging:** Do not reuse empty packaging. These may contain product residue and must be kept closed and sent for appropriate disposal as established for the product.

14. Transport information

National and international regulations

UN Number: 1170.
Proper shipping name: ETHANOL SOLUTION (ETHYL ALCOHOL SOLUTION)
Risk class/main risk subclass: 3
Risk class/subsidiary risk subclass: NA
Risk number: 33
Packing group: II

Waterway: DPC - Directorate of Ports and Coasts (Transportation in Brazilian waters)
Maritime Authority Regulations (NORMAM)
NORMAM 01/DPC: Vessels Used in Navigation on the Open Sea
NORMAM 02/DPC: Vessels Used in Inland Navigation
IMO – "International Maritime Organization" (Organização Marítima Internacional)

International Maritime Dangerous Goods Code (IMDG Code).

UN number: 1170.
Proper shipping name: ETHANOL SOLUTION (ETHYL ALCOHOL SOLUTION)
Primary risk class/subclass: 3
Subsidiary risk class/subclass: NA
Packaging group: II
EmS: F-E, S-D
Environmental hazard: The product is not considered a marine pollutant.
Air: ANAC - National Civil Aviation Agency – Resolution No. 129 of December 8, 2009.
RBAC No. 175 – (BRAZILIAN CIVIL AVIATION REGULATION) – TRANSPORTATION OF DANGEROUS ARTICLES IN CIVIL AIRCRAFT.
IS N° 175-001 – SUPPLEMENTARY INSTRUCTION - IS
ICAO – "International Civil Aviation Organization" – Doc 9284- NA/905
IATA - "International Air Transport Association"

Dangerous Goods Regulation (DGR).

UN number: 1170.
Proper shipping name: ETHANOL SOLUTION (ETHYL ALCOHOL SOLUTION)
Primary hazard class/subclass: 3
Subsidiary hazard class/subclass: NA
Packing group: II

15. Regulatory information

Federal Decree No. 2,657 of July 3, 1998 ABNT-NBR Standard 14725:2023.

Law No. 12,305 of August 2, 2010 (National Solid Waste Policy).

Decree No. 7,404 of December 23, 2010.

MTE Ordinance No. 704 of May 28, 2015 – Amends Regulatory Standard No. 26.

Product subject to control and inspection by the Ministry of Justice – Federal Police Department – MJ/DPF, in the case of import, export and re-export, with Prior Authorization from DPF being essential for carrying out these operations.

ANP Resolution 734/2018

Regulatory Standard No. 20

16. Other information

Important information, but not specifically described in the previous sections:

This MSDS was prepared based on current knowledge about the appropriate handling of the product and under normal conditions of use, according to the application specified on the packaging. Any other form of use of the product that involves its combination with other materials, in addition to forms of use other than those indicated, are the responsibility of the user. It is warned that the handling of any chemical substance requires prior knowledge of its hazards by the user. In the workplace, it is the responsibility of the company using the product to promote the training of its employees and contractors regarding the possible risks arising from exposure to the chemical product.

Captions and Abbreviations:

ACGIH - American Conference of Governmental Industrial Hygienists

BCF – Bioconcentration Factor CAS - Chemical Abstracts Service CL50 - Lethal concentration 50% LD50 - Lethal dose 50%

IARC - International Agency for Research on Cancer

LEI - Lower explosive limit LES - Upper explosive limit NA - Not applicable.

STEL - Short Term Exposure Limit

TLV - Threshold Limit Value

TWA - Time Weighted Average